

Production Order List — Basic Design Document (基本設計書)

Document control (改版履歴)

Field	Value
Document ID	002_BD
Category	UI
System name	ProductionManagementAI
Subsystem name	Production orders
Work item	WI-003
Based on brief.md revision	1
Created by	Claude (for ThanhTN)
Created date	2026-09-22
Last updated by	Claude (for ThanhTN)
Last updated date	2026-09-23

Version	Date	Author	Revision content
1	2026-09-22	Claude (for ThanhTN)	Initial creation (Screen B, production-order list; WI-003 DEC-001–DEC-009)
2	2026-09-22	Claude (for ThanhTN)	Reconciled with 002_DB: V-09 fragment bound clarified and \ added to the escaped characters (DEC-010); status sort defined as workflow order, not alphabetical
3	2026-09-22	Claude (for ThanhTN)	Message IDs renumbered to MSG-E015–MSG-E020 / MSG-I003–MSG-I004: the drafted IDs collided with 001_DD's catalog, which already defines MSG-E001–MSG-E014 and MSG-I001–MSG-I002; MSG-E002 and MSG-E013 reused instead of duplicated
4	2026-09-22	Claude (for ThanhTN)	WI-004 (DEC-016, DEC-021): the shared header now carries the application navbar specified in 003_BD "Shared application header"; items 1–2 point to it. The breadcrumb stays. No field, rule, validation, event or API of this screen changes
5	2026-09-23	Claude (for ThanhTN)	Diagrams redrawn for the revised BD template (RFC 0009): screen transition in Mermaid; PC and SP layouts as SVG wireframes under wireframes/. No field, rule, validation, event or API changes

Version	Date	Author	Revision content
6	2026-09-23	Claude (for ThanhTN)	WI-005: the UI is Japanese. Quoted UI text and messages give the implemented Japanese text with an English gloss (WI-005 DEC-003); status labels (M-05), result summary (M-09), date and timestamp formats (M-10, YYYY/MM/DD , YYYY/MM/DD HH:mm), page title and wireframes updated. No field, rule, validation, event or API changes

System overview

One screen, SCR-002, lets a signed-in Admin or Operator find production orders: a server-paged, sortable, filterable list of every order, from which an order is opened in SCR-001 (001_BD) for editing. It is "Screen B" on the locked roadmap and the screen that makes SCR-001 reachable by normal use instead of by a typed URL. SCR-002 is read-only: it changes no order data (DEC-003).

Requirement ID	Description	Covered by section
REQ-020	Open the list and see existing orders	Business flow; §3 items 14, 22, 24, 26, 28; §6 E-10; Success and exception flows
REQ-021	Row shows the order's identifying and planning fields	§3 items 16–22; §4 M-05–M-09
REQ-022	Filter by status, product, due-date range and order number	0-3 URL parameters; §3 items 7–13; §4 M-05, M-06; §5 V-09–V-12; §6 E-11, E-12
REQ-023	Sort by any listed column, ascending or descending	§3 items 16–21; §5 V-13; §6 E-13
REQ-024	Server-side paging with a user-chosen page size	§3 items 14, 15, 23; §5 V-13; §6 E-14, E-15
REQ-025	Open an order in SCR-001, or start a new one	Screen list and screen transition; §3 items 5, 22; §6 E-16, E-17
REQ-026	Only authenticated Admin/Operator	0-1 basic information; Actions and business rules; Exception flows
REQ-027	Filters, sort, page and page size kept in the URL	0-3 URL parameters; §6 E-10 to E-15

Overall configuration and architecture

Uses the confirmed stack (`ai/project.md`) exactly as 001_BD does: a React (Vite + TypeScript + Tailwind CSS v4) page calling the .NET 10 backend's JSON API over the same origin, layered Domain/Application/Infrastructure/Api (0001_ADR), persisted in PostgreSQL 17, gated by WI-001's cookie session and role checks (0002_ADR) — `ProtectedRoute` on the frontend, the same authorization policy on the backend. No new architecture boundary, external system or trust boundary is introduced.

SCR-002 adds one read-only query endpoint under the existing `/api/production-orders` path and reuses the existing product-list endpoint for its product filter. Every filter, sort key and page bound is resolved server-side; the browser never receives more than one page of rows. The exact contract is fixed in 002_DD-API;

index and query-plan support is fixed in 002_DB.

Function list

Function ID	Function name	Description	Related requirement ID
FN-010	Query production orders	Return one page of orders matching the current filters, in the requested order, with the total count of matches	REQ-020, REQ-022, REQ-023, REQ-024
FN-011	Validate query parameters	Reject an unknown status, unknown product, malformed or inverted date range, over-long order-number fragment, unsupported sort key/direction, or out-of-range page/page size	REQ-022, REQ-023, REQ-024
FN-012	Provide filter choices	Supply the product options (existing product list, FN-004) and the fixed status options for the filter panel	REQ-022
FN-013	Mark overdue orders	Flag a listed order whose due date is before the plant-timezone today and whose status is <code>Draft</code> or <code>InProgress</code>	REQ-021
FN-014	Keep view state in the URL	Read the view (filters, sort, page, page size) from the URL on load and write it back on every change	REQ-027
FN-015	Navigate to SCR-001	Open the selected order in edit mode, or open create mode from the New action	REQ-025
FN-016	Enforce authentication and role	Reject unauthenticated (401) and non-Admin/Operator (403) requests to the screen and its query endpoint	REQ-026

FN-004 (list products) is reused from 001_BD and is not redefined here.

Actors and business flow

Actors: Admin, Operator — both already signed in through WI-001's login screen (WI-002 DEC-001, carried into this screen unchanged).

Browse the list (UC-004)

1. User opens `/production-orders` from the home page → system reads the view state from the URL (empty on a plain link, FN-014), queries the first page with the default sort (FN-010) and loads the product filter choices (FN-012), showing a loading indicator until both finish.
2. System shows the page of rows, the range and total (`「87件中 1～20件」` , "1–20 of 87 orders"), and the active sort. No filter is pre-applied, so orders of every status are listed, including `Completed` and `Cancelled` (DEC-006).
3. If no order exists at all, the system shows the empty state with a `+ 新規製造指示` (New production order) action instead of an empty table.

Find an order (UC-005)

1. User sets any combination of status (multi-select, DEC-005), product, due-date range and order-number fragment, then presses **検索** (Search) (DEC-008) → system validates the filter values (FN-011); invalid values are shown as field errors and no query is sent.
2. System queries with the filters applied, returns to page 1 and updates the range, total and URL (FN-010, FN-014).
3. If nothing matches, the system shows the "no orders match" state with a **絞り込みをクリア** (Clear filters) action — not the unfiltered list.

Order and page through results (UC-006)

1. User clicks a column header → system sorts the whole result set by that column, returns to page 1 and marks the header with the direction; clicking the same header again reverses it (FN-010).
2. User changes the page size or moves between pages → system requests that page at that size; a page-size change returns to page 1 (FN-010).

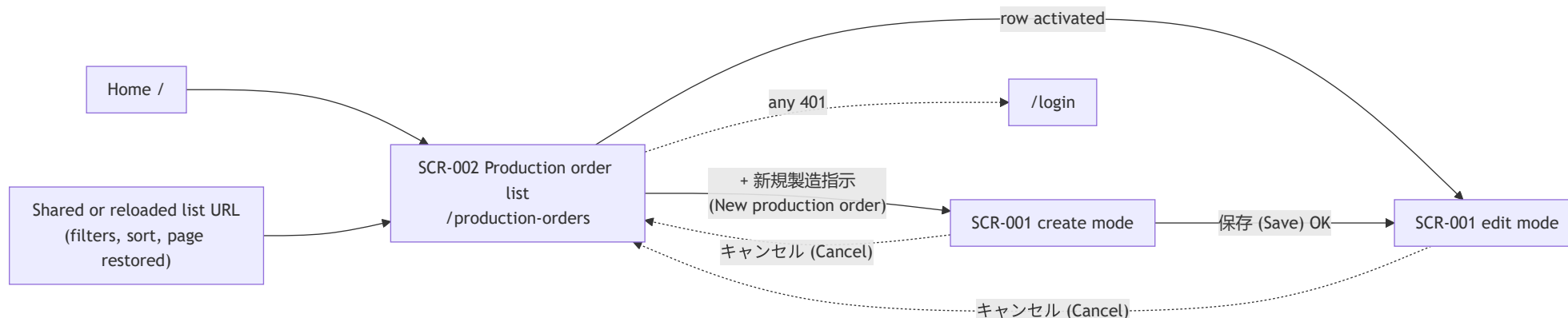
Act on an order (UC-007)

1. User activates a row (click, or the order-number link by keyboard) → system opens SCR-001 in edit mode for that order (FN-015).
2. User presses + **新規製造指示** (New production order) → system opens SCR-001 in create mode (FN-015).

Screen list and screen transition

Screen ID	Screen name	Entry point	Exit / next screen
SCR-002	Production Order List	<code>/production-orders</code> — a 「製造指示一覧」 (Production orders) link on the home page <code>/</code> (DEC-004). Also reached by Back from SCR-001, and by opening a shared list URL.	Row activated → SCR-001 edit mode <code>/production-orders/{id}</code> . 「+ 新規製造指示」 (New production order) → SCR-001 create mode <code>/production-orders/new</code> . Session expired / 401 → <code>/login</code> .

Screen transition:



SCR-001's Cancel currently returns to the home page `/` (001_BD, "until Screen B exists"). With SCR-002 in place, SCR-001's exit target becomes the list; that single change to 001_BD's screen transition is made when SCR-002 is implemented, and is the only Screen A behavior WI-003 touches.

The order status state machine is unchanged and is not restated here — see 001_BD, REQ-017. SCR-002 displays status but never changes it (DEC-003).

Screen design detail

SCR-002 Production Order List

0-1. Basic information (基本情報)

No	Item	Content	Reference
1	Route / path	<code>/production-orders</code>	0-3
2	API base path	<code>/api/production-orders</code> (query), <code>/api/products</code> (filter choices)	Exact contract in 002_DD-API
3	Character encoding	UTF-8	
4	Error page / fallback	Query failure: in-page error panel with a 再試行 (Retry) action, keeping the filter panel usable; no separate error route	Exception flows
5	Responsive	Yes — PC ($\geq 640\text{px}$) and SP ($< 640\text{px}$, Tailwind <code>sm</code> breakpoint)	§1

No	Item	Content	Reference
6	Authentication required	Yes	REQ-026
7	Authorization / role restriction	Admin , Operator (WI-002 DEC-001)	REQ-026
8	Applicable channel(s)	Single web app — not applicable	

0-2. Page metadata (head)

The app's common head (`src/frontend/index.html` : charset, viewport, favicon, bundled module script) is not restated.

0-2-1. Title

No	Title
1	製造指示一覧 – ProductionManagementAI (Production orders)

0-2-2. Base

Not applicable — no `<base>` override.

0-2-3. Link

Not applicable — no screen-specific `<link>` tags; styles are bundled.

0-2-4. Meta

Not applicable — no screen-specific meta tags (internal, authenticated screen; no SEO metadata needed).

0-2-5. Style

Not applicable — no inline `<style>` block; all styling via Tailwind classes.

0-2-6. Script

Not applicable — no extra `<script>` tags; app JS is bundled.

0-3. URL parameters

All parameters are query-string parameters on `/production-orders` ; every one is optional and the whole view state is reproducible from them (REQ-027). The frontend uses the same names as the API query (002_DD-API) so the two never drift.

No	Parameter name	Content	Required / Optional	Reference
1	status	Status filter; repeatable, one value per selected status (Draft , InProgress , Completed , Cancelled). Absent = no status restriction (DEC-005, DEC-006)	optional	§5 V-11
2	productId	Product filter; a single product identifier. Absent = all products	optional	§5 V-12
3	dueFrom	Start of the due-date range, inclusive, plant-local date YYYY-MM-DD	optional	§5 V-10
4	dueTo	End of the due-date range, inclusive, plant-local date YYYY-MM-DD	optional	§5 V-10
5	orderNumber	Case-insensitive fragment of the order number	optional	§5 V-09
6	sort	Sort key: orderNumber , product , quantity , dueDate , status , updatedAt . Absent = dueDate	optional	§5 V-13
7	dir	Sort direction asc or desc . Absent = asc	optional	§5 V-13
8	page	1-based page number. Absent = 1	optional	§5 V-13
9	pageSize	Rows per page; one of 10, 20, 50, 100. Absent = 20 (DEC-002)	optional	§5 V-13

An unreadable or out-of-range value in the URL falls back to that parameter's default and the screen still renders (REQ-027 failure criterion); it does not produce an error page. A rejected value sent directly to the API is a different case — see §5 V-09–V-13 and the exception flows.

1. Layout and mockup

Layout-level sketch only; the rendered per-state mockup belongs in 002_DD.

1 ProductionManagementAI

ダッシュボード

製造指示一覧

+ 新規製造指示

2 Admin・ログアウト

3 ホーム > 製造指示一覧

4 製造指示一覧

5 + 新規製造指示

6 message banner: query error · 再試行

7 ステータス

☐ 下書き ☐ 進行中 ☐ 完了 ☐ 取消

8 製品

すべての製品

9 納期（開始）

yyyy/mm/dd

10 納期（終了）

yyyy/mm/dd

11 指示番号

PO-2026-...

12 クリア

13 検索

14 87件中 1～20件

15 表示件数

20

16 指示番号 ▲	17 製品	18 数量	19 納期	20 ステータス	21 更新日時
22 PO-2026-00001	P-1001 — プレーキキャリパー	120	2026/09/30	下書き	2026/09/20 10:12
PO-2026-00002	P-1004 — ドライブシャフト	40	2026/09/18	納期遅れ 27 進行中	2026/09/19 16:40
PO-2026-00003	P-1006 — トランスミッションケース	240	2026/10/02	下書き	2026/09/19 11:05
...					

23 < 前へ

1 / 5 ページ

次へ >

Not drawn: 24 empty state, 25 no-match state, 26 loading indicator (they replace the table body).

Item No.	Region / element	Notes (behavior, condition)
1	App header with navbar	Shared application header, specified once in 003_BD "Shared application header" (H-1–H-5, E-29–E-31, WI-004 DEC-016): app name, navbar (「ダッシュボード」 (Dashboard), 「製造指示一覧」 (Production orders), 「新規製造指示」 (New production order); current entry marked), user and 「ログアウト」 (Sign out), a 「メニュー」 (Menu) button on SP. Current entry here: 「製造指示一覧」
2	Signed-in user + 「ログアウト」 (Sign out)	Shared header (003_BD H-3; WI-001 behavior unchanged)
3	Breadcrumb	「ホーム > 製造指示一覧」 (Home > Production orders)
4	Page heading	「製造指示一覧」 (Production orders)
5	「+ 新規製造指示」 (New production order)	Primary action, top right; opens SCR-001 create mode
6	Message banner	Hidden unless the query failed; carries the 再試行 (Retry) action; announced to assistive tech
7–13	Filter panel	See §3; one card above the results, collapsible on SP
14	Result summary	「{total}件中 {first}～{last}件」 ({first}–{last} of {total} orders); the live region that announces a finished query
15	Page-size select	10 / 20 / 50 / 100, default 20
16–21	Sortable column headers	Each a button; the active one shows ▲/▼ and aria-sort
22	Result row	One per order; the order number is a link to SCR-001 edit mode (DEC-009)
23	Paging controls	「< 前へ」 / 「次へ >」 (Previous / Next) and 「{n} / {m} ページ」 (Page {n} of {m}); disabled at the ends
24	Empty state	Replaces 14–23 when the system holds no orders at all: message MSG-I003 and a + 新規製造指示 (New production order) action
25	No-match state	Replaces the table body when filters match nothing: message MSG-I004 and a 絞り込みをクリア (Clear filters) action; the filter panel stays filled in
26	Loading indicator	「製造指示を読み込み中…」 (Loading production orders...), shown in place of the table body while a query is in flight; the previous rows are not left on screen as if they were the result
27	Overdue marker	「納期遅れ」 (Overdue), shown next to the due date of an overdue row (§4 M-08)

SP / mobile

1 PMAI

2 メニュー

3 < ホーム

4 製造指示一覧

5 + 新規製造指示

6 message banner

7-13 絞り込み (2) ▼

14 87件中 1～20件

15 表示件数 20 ▼

16 並び替え 納期 ▲ ▼

22 **PO-2026-00001**
P-1001 — プレーキキャリパー
数量 120
納期 2026/09/30
下書き・更新 2026/09/20 10:12

PO-2026-00002
P-1004 — ドライブシャフト
数量 40
納期 2026/09/18
進行中・更新 2026/09/19 16:40

... one card per order

23 < 前へ 1 / 5 ページ 次へ >

Same item numbers as PC. Differences: the table becomes one card per order, each card as a whole being the link to SCR-001; the filter panel collapses behind a 絞り

り込み (Filters) disclosure showing how many filters are active; the sortable column headers collapse into a single sort select (item 16) offering the same keys and directions as items 16–21.

2. Content block definition (CMS)

None — no externally managed content blocks.

3. Screen item definition

No	Item (label)	Variable name	Control type	I/O	Data type	Width / length	Initial value	Placeholder	Display condition	Data source	Reference
4	Page heading	—	label	O	text	—	「製造指示一覧」 (Production orders)	—	Always	—	
5	「+ 新規製造指示」 (New production order)	—	button (link)	I	—	—	—	—	Always	—	\$6 E-17
6	Message banner	—	label	O	text	—	Hidden	—	On query failure	Query result	\$6 E-18
7	「ステータス」 (Status) filter	status	checkbox group	I	enum list	4 options	None checked (DEC-006)	—	Always	Fixed status list	\$4 M-05; DEC-005
8	「製品」 (Product) filter	productId	select	I	identifier	—	「すべての製品」 (All products)	—	Always	Product list (FN-004, FN-012)	\$4 M-06
9	「納期 (開始)」 (Due from)	dueFrom	date picker	I	date	—	Empty	—	Always	—	\$5 V-10
10	「納期 (終了)」 (Due to)	dueTo	date picker	I	date	—	Empty	—	Always	—	\$5 V-10
11	「指示番号」 (Order number)	orderNumber	textbox	I	text	≤ 20 chars	Empty	"PO-2026-..."	Always	—	\$5 V-09

No	Item (label)	Variable name	Control type	I/O	Data type	Width / length	Initial value	Placeholder	Display condition	Data source	Reference
12	「クリア」 (Clear)	—	button	I	—	—	—	—	Always; disabled when no filter is set	—	\$6 E-12
13	「検索」 (Search)	—	button (submit)	I	—	—	—	—	Always; disabled while a query is in flight	—	\$6 E-11
14	Result summary	—	label (live region)	O	text	—	—	—	Whenever rows are shown	Query result total , page , pageSize	\$4 M-09
15	「表示件数」 (Rows per page)	pageSize	select	I	integer	—	20	—	Whenever rows are shown	Fixed list 10/20/50/100	\$5 V-13; \$6 E-15
16	「指示番号」 (Order no.) header	sort=orderNumber	button (column header)	I	—	—	—	—	PC; on SP an option of item 16's sort select	—	\$6 E-13
17	「製品」 (Product) header	sort=product	button (column header)	I	—	—	—	—	as item 16	—	Sorts by SKU then name
18	「数量」 (Qty) header	sort=quantity	button (column header)	I	—	—	—	—	as item 16	—	
19	「納期」 (Due date) header	sort=dueDate	button (column header)	I	—	—	Active by default, ascending	—	as item 16	—	Default sort (DEC-002)
20	「ステータ ス」 (Status) header	sort=status	button (column header)	I	—	—	—	—	as item 16	—	\$4 M-07; sorts in workflow order (下書き, 進行中, 完了, 取消 — Draft, In progress, Completed, Cancelled), not alphabetically — 002_DB sort-key mapping
21	「更新日時」 (Updated) header	sort=updatedAt	button (column header)	I	—	—	—	—	as item 16	—	

No	Item (label)	Variable name	Control type	I/O	Data type	Width / length	Initial value	Placeholder	Display condition	Data source	Reference
22	Result row	—	table row (order number is a link)	O	—	—	—	—	One per returned order	Query result rows	\$4 M-05–M-08; \$6 E-16
23	Paging controls	page	buttons + label	I/O	integer	—	Page 1	—	Whenever rows are shown	Query result total , page , pageSize	\$6 E-14
24	Empty state	—	label + button	O	text	—	Hidden	—	Only when the system holds no orders at all	Query result (total = 0, no filter set)	MSG-I003
25	No-match state	—	label + button	O	text	—	Hidden	—	Only when filters are set and match nothing	Query result (total = 0, filter set)	MSG-I004; \$6 E-12
26	Loading indicator	—	label	O	text	—	Hidden	—	While a query is in flight	—	\$6 E-10
27	Overdue marker	—	label	O	text	—	Hidden	—	Per row, per M-08	Row due date + status + plant today	\$4 M-08

Every row shows, in order: order number (item 22's link text), product, quantity, due date (with the overdue marker where applicable), status and last updated (REQ-021).

4. Item value mapping

No	Item	Source value	Displayed value	Reference
M-05	(7) Status filter options / (22) row status	Draft / InProgress / Completed / Cancelled	「下書き」 / 「進行中」 / 「完了」 / 「取消」 (Draft / In progress / Completed / Cancelled)	Same labels as 001_BD M-01
M-06	(8) Product filter / (22) row product	Product ID	"{SKU} — {name}", ordered by SKU; the filter's first option is 「すべての製品」 (All products) with an empty value	001_BD M-03, WI-002 DEC-022

No	Item	Source value	Displayed value	Reference
M-07	(22) Row status	Status value	Status label (M-05) rendered as a badge whose styling distinguishes active (<code>Draft</code> , <code>InProgress</code>) from terminal (<code>Completed</code> , <code>Cancelled</code>) states; the label alone carries the meaning, never color alone	WCAG 1.4.1
M-08	(27) Overdue marker	Row due date, row status, plant-local today	Marked 「納期遅れ」 (Overdue) when due date < today in Asia/Tokyo (WI-002 DEC-011, DEC-017) and status is <code>Draft</code> or <code>InProgress</code> ; never for <code>Completed</code> or <code>Cancelled</code>	REQ-021; FN-013
M-09	(14) Result summary	<code>page</code> , <code>pageSize</code> , <code>total</code>	「{total}件中 {first}～{last}件」, where first = (page-1)×pageSize + 1 and last = min(page×pageSize, total) ({first}–{last} of {total} orders); with <code>total</code> = 0 the summary is replaced by item 24 or 25	REQ-024
M-10	(22) Row updated / due date	Stored timestamp (UTC) / stored date	Updated: YYYY/MM/DD HH:mm , 24-hour, in the browser's timezone (as 001_BD M-04). Due date: the stored plant-local date as YYYY/MM/DD , with no timezone conversion	WI-005 DEC-006

5. Validation rules

The filter panel checks values client-side for early feedback; the server repeats every check on the query endpoint and is authoritative. A value rejected by the server returns an error rather than silently falling back to a default, so a crafted request cannot widen the query (REQ-022–REQ-024 failure criteria). Error message texts are drafts; final IDs and wording are fixed in 002_DD.

No	Item	Check content	Validation rule	Check condition	Error message (ID)	Reference
V-09	(11) Order number	Length and content of the search fragment	≤ 20 characters after trimming (a bound, not a format check — a full order number is 13 characters); matched case-insensitively as a fragment of the order number; the wildcard characters <code>%</code> , <code>_</code> and <code>\</code> are treated as literal text, not as patterns	On Search; on query	MSG-E015 「指示番号の検索は20文字以内で入力してください。」 (Order number search can't exceed 20 characters.)	REQ-022; 002_DD-API fixes the escaping
V-10	(9)(10) Due from / Due to	Valid dates, ordered range	Each a valid date; if both are set, <code>dueFrom</code> ≤ <code>dueTo</code> ; range inclusive at both ends; either end may be left empty	On Search; on query	MSG-E016 「有効な日付を入力してください。」 (Enter a valid date.) / MSG-E017 「納期（開始）は納期（終了）以前の日付にしてください。」 ('Due from' must be on or before 'Due to'.)	REQ-022
V-11	(7) Status filter	Known status values	Every submitted value is one of <code>Draft</code> , <code>InProgress</code> , <code>Completed</code> , <code>Cancelled</code> ; none submitted = no restriction; duplicates collapse to one	On query (server)	MSG-E018 「不明なステータスです。」 (Unknown order status.)	REQ-022, DEC-005
V-12	(8) Product filter	Existing product	The submitted product identifier exists in Product	On query (server)	MSG-E002 「選択した製品は存在しません。」 (The selected product no longer exists.) (reused from 001_DD — same meaning, same wording)	REQ-022; 001_BD V-01

No	Item	Check content	Validation rule	Check condition	Error message (ID)	Reference
V-13	(15)(16–21) (23) Paging and sorting	Bounded, allow-listed query controls	<code>sort</code> ∈ {orderNumber, product, quantity, dueDate, status, updatedAt}; <code>dir</code> ∈ {asc, desc}; <code>page</code> an integer ≥ 1; <code>pageSize</code> ∈ {10, 20, 50, 100}. Rejected — not defaulted — when sent explicitly with an invalid value	On query (server)	MSG-E019 「対応していない並べ替えまたはページングの指定です。」 (Unsupported sort or paging option.)	REQ-023, REQ-024; DEC-002

Message IDs continue 001_DD's single catalog (`src/frontend/src/features/production-orders/messages.ts`), which already uses MSG-E001–MSG-E014 and MSG-I001–MSG-I002. Screen B's new IDs therefore start at MSG-E015 and MSG-I003; where 001_DD already has a message with the same meaning, it is reused rather than duplicated.

ID	Text (draft)	Shown when	New or reused
MSG-E015	「指示番号の検索は20文字以内で入力してください。」 (Order number search can't exceed 20 characters.)	V-09 fails	new
MSG-E016	「有効な日付を入力してください。」 (Enter a valid date.)	V-10, malformed date	new
MSG-E017	「納期（開始）は納期（終了）以前の日付にしてください。」 ('Due from' must be on or before 'Due to'.)	V-10, inverted range	new
MSG-E018	「不明なステータスです。」 (Unknown order status.)	V-11 fails (crafted request)	new
MSG-E019	「対応していない並べ替えまたはページングの指定です。」 (Unsupported sort or paging option.)	V-13 fails (crafted request)	new
MSG-E020	「製造指示を閲覧する権限がありません。」 (You don't have permission to view production orders.)	403 on the list	new — MSG-E012 says "manage", which is wrong for a read-only screen
MSG-E002	「選択した製品は存在しません。」 (The selected product no longer exists.)	V-12 fails	reused from 001_DD
MSG-E013	「問題が発生しました。もう一度お試しください。」 (Something went wrong. Try again.)	Query/network failure (item 6)	reused from 001_DD
MSG-I003	「製造指示はまだありません。最初の1件を登録してください。」 (No production orders yet. Create the first one.)	The system holds no orders at all (item 24)	new
MSG-I004	「絞り込みに一致する製造指示はありません。」 (No orders match your filters.)	Filters are set and match nothing (item 25)	new

6. Item events

No	Item	Event	Event content	Reference
E-10	Screen	load	Read the view state from the URL (0-3), falling back per parameter to its default; load the product filter choices (FN-012) and run the query (FN-010); show item 26 while in flight; on 401 go to <code>/login</code> ; on failure show item 6 with 再試行 (Retry)	FN-010, FN-012, FN-014
E-11	(13) Search	click / Enter in the filter panel	Run the client-side filter checks (V-09, V-10); if any fail, show inline errors, keep the previous rows and send no query. Otherwise return to page 1, keep the current sort and page size, run the query and write the new view state to the URL as a new history entry	REQ-022, REQ-027; DEC-008
E-12	(12) 「クリア」 (Clear) / (25) 「絞り込みをクリア」 (Clear filters)	click	Reset items 7–11 to their initial values, return to page 1, keep sort and page size, query and update the URL	REQ-022
E-13	(16–21) Column header	click / Enter / Space	If the header is already the active sort key, reverse <code>dir</code> ; otherwise make it the sort key with <code>asc</code> . Return to page 1, query, update the URL and <code>aria-sort</code>	REQ-023
E-14	(23) 「< 前へ」 / 「次へ >」 (Previous / Next)	click	Move one page within 1...last page; the control is disabled at either end. Query and update the URL	REQ-024
E-15	(15) Page size	change	Apply the new page size, return to page 1, query and update the URL	REQ-024, DEC-002
E-16	(22) Result row	click (row) / Enter on the order-number link	Open <code>/production-orders/{id}</code> (SCR-001 edit mode). The order-number cell is a real link, so the row is reachable and activatable by keyboard; the whole-row click is a mouse convenience that resolves to the same link and never fires on a text selection (DEC-009)	REQ-025, FN-015
E-17	(5) 「+ 新規製造指示」 (New production order)	click	Open <code>/production-orders/new</code> (SCR-001 create mode)	REQ-025, FN-015
E-18	(6) 「再試行」 (Retry)	click	Re-run the last query unchanged	Exception flows
E-19	Browser Back / Forward, or a pasted list URL	popstate / load	Re-read the view state from the URL and re-run the query, so history navigation reproduces the earlier view	REQ-027

No event on this screen writes order data (DEC-003).

7. External identity linkage

None — no external identity linkage.

Actions and business rules

Action	Trigger	Business rule	Related requirement ID
Open screen / call the query API	Any request to SCR-002 or its query endpoint	Requires a valid session (else 401 → <code>/login</code>) and role <code>Admin</code> or <code>Operator</code> (else 403) — enforced server-side, not only in the UI	REQ-026
Query orders	Screen load, Search, Clear, sort, page, page-size change	Filters combine with AND; an absent filter places no restriction; the total reported is the number of orders matching the same filters, independent of the page	REQ-020, REQ-022
Sort results	Column header (item 16–21) or SP sort select	Only the six allow-listed keys, ascending or descending, applied to the whole result set before paging; order number ascending is the tie-breaker so paging never repeats or skips a row	REQ-023
Page results	Paging controls, page-size select	Every query is bounded by the page size; the page size is one of the four allowed values; a page beyond the last returns no rows rather than an error	REQ-024
Open an order	Row activation	Navigation only — the list reads order data and never writes it	REQ-025, DEC-003
Mark overdue	Row rendering	Due date < plant-local today and status ∈ { <code>Draft</code> , <code>InProgress</code> }	REQ-021

Success and exception flows

Flow	Trigger condition	System behavior	Resulting state
Success — first load	Authorized user opens the screen with no filters	First page returned in the default sort with the total	Table with rows, summary, sort marker, paging controls
Success — filtered	Valid filters submitted	Matching page and total returned, page reset to 1, URL updated	Table with the filtered rows; filter panel keeps its values
Success — sorted / paged	Sort, page or page-size change	Requested slice returned	Table updated; summary, sort marker and URL agree with it
Success — restored view	A shared or reloaded list URL, or Back/Forward	View state read from the URL and re-queried	Same view as when the URL was produced
Empty — no orders at all	The system holds no production orders	Empty state instead of a table	MSG-I003 with + 新規製造指示 (New production order)
Empty — no match	Filters set, nothing matches	No-match state instead of rows	MSG-I004 with 絞り込みをクリア (Clear filters); filter values kept
Exception — unauthenticated	No valid session (including expiry while browsing)	401, no data returned	Redirect to <code>/login</code>

Flow	Trigger condition	System behavior	Resulting state
Exception — forbidden	Signed in without Admin / Operator	403, no data returned	Error panel MSG-E020 「製造指示を閲覧する権限がありません。」 (You don't have permission to view production orders.)
Exception — invalid filter value	V-09–V-12 fails	Query rejected; no rows replaced by wrong ones	Inline field error (client) or error banner (crafted request); previous rows stay
Exception — invalid sort/paging option	V-13 fails on an explicit value	Query rejected rather than defaulted	Error banner; previous rows stay
Exception — page beyond the last	page > last page for the current filters	Query succeeds with no rows	Empty page with paging controls intact; no error
Exception — product choices unavailable	The product list fails to load	The product filter is disabled and shows a load error; every other filter still works	Rest of the screen usable
Exception — query/server/network error	Unexpected failure	Nothing shown as a result	Error banner with 再試行 (Retry); filter values kept

Data design overview

No new entity. The UI is Japanese only (WI-005 DEC-001); product names are Japanese automobile parts (WI-005 DEC-002). SCR-002 reads the entities 001_BD already defined: **ProductionOrder** (order number, product reference, quantity, due date, status, notes, created/updated timestamps) joined to its **Product** (SKU, name) for the displayed and sortable product column. Notes are not shown in the list.

What database design has to add (002_DB), rather than restate:

- Index support for the default sort (due date, order number) and for each filter — status, product, due-date range and the case-insensitive order-number fragment — so the list stays index-backed as the table grows.
- The count query that produces the total alongside the page.
- The demo seed of roughly 60–100 production orders across statuses, products and due dates (DEC-007), including past due dates so the overdue marker is demonstrable.

Column-level detail, index definitions and the seed's environment scope belong to 002_DB.

External interfaces

None — only this application's own backend API.

Non-functional requirements

- **Security:** the role check (Admin / Operator) gates the query endpoint server-side, not only in the UI (REQ-026). The sort key, sort direction and page size are allow-listed rather than interpolated, so no client value reaches the query as SQL; the page size is capped at 100 so no request can pull an unbounded result set; every filter value is bound as a parameter, and % and _ in the order-number fragment are escaped so a search cannot widen itself into a full scan. No field shown on this screen is PII or a secret. CSRF is not a concern for this read-only GET endpoint, and the WI-002 posture is unchanged (WI-002 DEC-020).
- **Accessibility (WCAG 2.2 AA, ai/rules/frontend.md):** the results are a real <table> with a caption and header cells; sortable headers are buttons carrying aria-sort on the active column; every row is reachable by keyboard through its order-number link, so the mouse-only whole-row click adds no keyboard-inaccessible behavior (DEC-009); the result summary is a live region that announces the new total when a query finishes; the loading state is announced rather than shown only as a spinner; status badges and the overdue marker carry text, never color alone; filter fields have visible labels, inline errors linked by aria-describedby , and focus moves to the first invalid filter when Search fails; paging controls state their target (「前のページ」 "Previous page", 「次のページ」 "Next page") and are disabled — not hidden — at the ends; visible focus indicators and sufficient contrast throughout.
- **Performance:** every query returns at most one page of rows plus a count; both are index-backed per 002_DB. The demo target is a first page rendered well within a second on the seeded data set; no client-side filtering or sorting of a full result set is performed.
- **Observability:** the query endpoint is traced and logged per project defaults, as 001_BD's endpoints are; exact spans and metrics are specified in 002_DD-FN.
- **Availability:** inherits project defaults.

Open questions and linked DD

Question	Linked DD section	Status
Status filter single- or multi-select	002_DD-API query contract	answered in decisions.md (DEC-005) — multi-select
Whether the default view hides terminal statuses	002_DD initial state	answered in decisions.md (DEC-006) — shows all
Whether filters apply instantly or on an explicit Search	002_DD events	answered in decisions.md (DEC-008) — explicit Search

Question	Linked DD section	Status
How a row is activated without breaking keyboard access	002_DD events / accessibility	answered in decisions.md (DEC-009) — order-number link plus mouse row click
Exact wording and IDs of the new messages	002_DD message list	answered — renumbered to MSG-E015–MSG-E020 and MSG-I003–MSG-I004 in version 3, after 001_DD's catalog was checked; final wording confirmed in 002_DD
Order-number fragment matching: escaping and index strategy	002_DD-API; 002_DB	open — 002_DB decides the index; 002_DD-API fixes the escaping
Seeded demo orders: count, spread and environment scope	002_DB seed section	answered in principle (DEC-007); detail fixed in 002_DB